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Smart Agriculture System

train_model.py | predict.py | farm_dataset.csv | AI_Brain.py | requirements.txt | smart_agriculture.ino

EXPLORER

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OUTLINE

TIMELINE

AI-driven-agriculture-project > dataset > farm_dataset.csv > data

```csv
temp,humidity,light,soil_moisture,next_soil
24,33,92,88,84.7
21,55,94,70,67.23
33,69,58,51,48.99
36,79,30,76,74.88
31,54,70,71,68.43
22,39,29,73,71.81
30,54,27,62,60.77
25,49,39,66,64.56
28,43,45,62,60.11
31,33,46,40,37.730000000000004
30,77,45,83,81.69
24,76,40,53,52.12
32,47,75,72,69.09
20,58,37,79,78.05
32,31,40,42,39.82
34,79,97,66,62.97
27,75,93,44,41.36
25,54,44,67,65.51
40,35,40,60,57.5
37,69,36,73,71.45
36,36,32,78,75.96
35,42,54,47,44.47
27,33,94,52,48.49
31,53,85,71,67.96
24,58,91,45,42.23
25,72,73,86,84.0
24,56,89,59,56.25
35,73,68,40,37.67
37,43,93,49,45.22
35,77,74,58,55.57
34,64,55,40,37.93
36,63,50,61,58.96
26,42,40,55,53.34
34,37,48,88,85.6
23,58,22,43,42.35
27,70,20,75,74.45
```

main | 0 0 | CSVLint | Query | Align

Col 1: temp | Ln 1, Col 1 | Spaces: 4 | UTF-8 | CRLF | CSV | Finish Setup | Go Live

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Smart Agriculture System

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AI-driven-agriculture-project > ai_server > train_model.py > ...

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score
import joblib
import os

BASE_DIR = os.path.dirname(__file__)

# correct dataset path
dataset_path = os.path.join(BASE_DIR, "..", "dataset", "farm_dataset.csv")

data = pd.read_csv(dataset_path)

X = data[['temp','humidity','light','soil_moisture']]
y = data['next_soil']

# split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# train model
model = RandomForestRegressor(n_estimators=200, random_state=42)
model.fit(X_train, y_train)

# test accuracy
predictions = model.predict(X_test)
accuracy = r2_score(y_test, predictions)

print("Model Accuracy:", accuracy)

# save model in ai_server folder
model_path = os.path.join(BASE_DIR, "soil_model.pkl")
joblib.dump(model, model_path)

print("Model saved at:", model_path)
```

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```python
import joblib
import pandas as pd

model = joblib.load("soil_model.pkl")

temp = 31
humidity = 50
light = 80
soil = 64

input_data = pd.DataFrame([[temp,humidity,light,soil]],
columns=['temp','humidity','light','soil_moisture'])

prediction = model.predict(input_data)

print("Predicted next soil moisture:", prediction[0])

dry_threshold = 35
drop_rate = soil - prediction[0]

hours_to_dry = (soil - dry_threshold) / drop_rate

print("\nSoil Moisture:",soil,"%")
print("Temperature:",temp,"°C")

print("\nPrediction:")
print("Soil will dry in",round(hours_to_dry,2),"hours")

def score_range(value, low, high):

    if low <= value <= high:
        return 100
    elif value < low:
        return max(0, 100 - (low - value) * 5)
    else:
        return max(0, 100 - (value - high) * 5)
```

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OUTLINE

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AI-driven-agriculture-project > ai_server > AI_Brain.py > ...

```python
from flask import Flask, request, jsonify
import pandas as pd
import joblib
import requests
import os

app = Flask(__name__)

# load trained model
BASE_DIR = os.path.dirname(__file__)
model_path = os.path.join(BASE_DIR, "soil_model.pkl")
model = joblib.load(model_path)

# get token from environment variable
BLYNK_TOKEN = os.getenv("BLYNK_TOKEN")
if BLYNK_TOKEN is None:
    print("WARNING: BLYNK TOKEN NOT FOUND")


def farm_health(soil, temp, humidity, light):

    def score_range(value, low, high):
        if low <= value <= high:
            return 100
        elif value < low:
            return max(0, 100 - (low - value) * 5)
        else:
            return max(0, 100 - (value - high) * 5)

    soil_score = score_range(soil, 50, 70)
    temp_score = score_range(temp, 20, 32)
    humidity_score = score_range(humidity, 40, 70)
    light_score = score_range(light, 30, 80)

    health = (
        soil_score * 0.4 +
        temp_score * 0.25 +
```

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EXPLORER

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AI-driven-agriculture-project > ai_server > AI_Brain.py > ...

```python
from flask import Flask, request, jsonify
import pandas as pd
import joblib
import requests

app = Flask(__name__)

# load trained model
model = joblib.load("soil_model.pkl")

BLYNK_TOKEN = "IWvwGsrSqTh8I02Z-3LHe0K7MNd8YA3U"


def farm_health(soil, temp, humidity, light):

    def score_range(value, low, high):
        if low <= value <= high:
            return 100
        elif value < low:
            return max(0, 100 - (low - value) * 5)
        else:
            return max(0, 100 - (value - high) * 5)
```

PROBLEMS 5 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

powershell - AI-driven-agriculture-project

```
PS C:\Users\Dell\OneDrive\ドキュメント\Smart Agriculture System\AI-driven-agriculture-project> git status
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   ai_server/AI_Brain.py
        modified:   ai_server/predict.py
        modified:   ai_server/requirements.txt
        modified:   ai_server/soil_model.pkl
        modified:   ai_server/train_model.py

PS C:\Users\Dell\OneDrive\ドキュメント\Smart Agriculture System\AI-driven-agriculture-project>
```

OUTLINE

TIMELINE

dashboard.render.com/web/srv-d6mlthtm5p6s73ftloeg

Aryan's workspace | Smart-Agriculture-AI | Search K | + New | Upgrade

← Dashboard

Smart-Agriculture-AI

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MANAGE

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Changelog

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Render Status

WEB SERVICE

Smart-Agriculture-AI

Python 3 | Free | Upgrade your instance →

Connect | Manual Deploy

Service ID: srv-d6mlthtm5p6s73ftloeg

sisodia91 / AI-driven-agriculture-project main

https://smart-agriculture-ai-2t3f.onrender.com

Your free instance will spin down with inactivity, which can delay requests by 50 seconds or more. Upgrade now

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Deploy live for 92f6ef1: Fix Blynk update and indentation
March 8, 2026 at 6:13 PM

Deploy started for 92f6ef1: Fix Blynk update and indentation
Manually triggered by you via Dashboard
March 8, 2026 at 6:05 PM

Deploy live for 92f6ef1: Fix Blynk update and indentation
March 8, 2026 at 6:05 PM
Rollback